

Deterministic 3-Tier Conformal Risk Control (CRC) Operational Triage Pipeline

Incoming Stream

$N = 10,000$ tx

- Live Event Buffer
- High-Throughput
- Zero-Copy Arrow
- Dynamic Scoring

C-STGB Backbone

Non-Conformity s_i

- $s_i^{(y)} = 1 - \hat{p}_i(y)$
- Dual Non-Conf. Scores
 - Bayes Policy τ^*
 - Streaming 0.45 ms

Class-Conditional CRC

$$\Gamma(X) = \{y : s_i^{(y)} \leq \hat{q}^{(y)}\}$$

- Dual Quantiles:
 $\hat{q}^{(0)}$ (Benign), $\hat{q}^{(1)}$ (Illicit)
- Coverage Guarantee:
 $P(Y \in \Gamma(X) \mid y) \geq 99\%$
- Drift Robustness:
DKW Bound $+\sqrt{|\mathcal{D}_{\text{cal}}|}$

Tier 1: Instant Auto-Block

$$\Gamma(X) = \{1\}$$

- Volume: $\approx 55 / 10k$ tx (0.55%)
- Action: Precision > 99.4% | Freeze & SAR Draft

Tier 2: Manual Review Queue

$$\Gamma(X) = \{0, 1\}$$

- Volume: $\approx 5 / 10k$ tx (0.05% Ambiguous)
- Action: Dispatches Multi-Agent Forensic Dossier

Tier 3: Straight-Through Auto-Clear

$$\Gamma(X) = \{0\}$$

- Volume: $\approx 9,940 / 10k$ tx (99.40%)
- Action: Benign Guarantee > 99.9% | Instant STP

>99.4% Net Compliance Workload Reduction (Human Review Focused Exclusively on the 0.05% Ambiguous Queue)